Guilherme Ruiz

PhD in Particle Physics

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Resume PDF

CAREER PROFILE

Particle physicist with a PhD from the University of Manchester, specialising in solar neutrino detection using liquid argon time projection chambers. MSc, University College London; BSc, University of São Paulo. Experience spanning experimental physics, data analysis, and software development, including professional work in operational strategy at Amazon. Licensed private pilot in both Brazil and the United Kingdom.

WORK EXPERIENCES

Network Control Manager 2020 - 2021
Amazon Logistics, London

Started off as an intern and had a position specially created for me to be hired full time.

- Automation of reports and data processing through Python scripts, SQL and Excel
- Managing multiple stakeholders and customers, presenting data and promptly handling last minute requests
- Prioritisation of workload between long term strategic projects and short term tactical projects

Undergraduate Research Assistant 2017
University of São Paulo

Programatically (Python) collected, organised and thoroughly analysed a large volume of data from numerous sources to determine the mechanism behind η Carinae's brightening in the last century.

- Data analysis and interpretation of photometric data
- Development of a Python code to analyse the data

Undergraduate Researcher Assistant 2014 - 2016
University of São Paulo

Assisted the construction and tests of GEM type detectors related to the upgrade of the ALICE experiment at the LHC.

- Funded by CNPq - PIBIC scholarship
- Presented at the 23rd (2015) and 24th (2016) International Symposium on Scientific Initiation of the University of São Paulo (SIICUSP)

Volunteer 2014 - 2016
Institute of Physics' LGBT+ Collective

A founding member in 2014. As one of four direction officers from 2015 to 2016:

- Planned and coordinated the creation of the collective's manifesto
- Managed social media and events
- Persuaded the institute's administration into implementing new policies for guaranteeing the wellbeing of transgender students
- Held many joint meetings with the student's union officers

EVENTS

Speaker — IOP Joint APP, HEPP and NP Annual Conference 2024 2024
The Spire, Liverpool, UK

Presentation on the SoLAr detector prototyping campaign.

Speaker — NuPhys2023 2023
Kings College London, UK

Brief presentation and poster on the SoLAr concept for solar neutrino detection.

Volunteer — Particle Physics Masterclass 2023
University of Manchester, UK

Science outreach event for sixth form students and their teachers.

- Served as a member of the outreach team
- Presented the department's laboratory facilities
- Explained the department's work in an accessible way

Contestant — UCL Data Science Society Hackathon 2019
Microsoft Reactor, London, UK

Data science competition in partnership with American Express and Microsoft.

- Explored the use of data science to address challenges in the credit card industry
- Teamwork and collaboration under time constraints

Speaker — 24th SIICUSP 2016
University of São Paulo, Brazil

Brief presentation of research on GEM detectors for the ALICE experiment at LHC.

Speaker — 23rd SIICUSP 2015
University of São Paulo, Brazil

Brief presentation of research on GEM detectors for the ALICE experiment at LHC.

Volunteer — Science Fair 2014
University of São Paulo, Brazil

Science outreach event for the general public.

- Served as staff for the department of nuclear physics
- Guided groups of students through experiments and activities
- Explained scientific concepts in an accessible way

CERTIFICATIONS

Private Pilot License 2018
Brazilian Civil Aviation Authority (349370) ANAC CHT

Brazilian private pilot license for single engine piston aircraft.

Private Pilot License 2024
UK Civil Aviation Authority (553115B) CAA PPL

British private pilot license for single engine piston aircraft.

PROJECTS

Personal projects and academic assignments.

SoLAr V2 reco - Reconstruction and data analysis software for the SoLAr experiment's second prototype detector.

YugiQuery - Python script to query and display Yu-Gi-Oh! data from the yugipedia database.

Arduino Odometry - Jupyter notebook to reconstruct inertial navigation data from arduino sensors.

nbtools - Collection of Jupyter notebook utilities to improve productivity and usability.

DHT Weather Library - Arduino library for reading the DHT family of sensors and performing common weather calculations.

OSS CONTRIBUTIONS

Pull requests and contributions to open source projects.

TQDM Discord contrib fix - Implementation of Discord Rich Presence for the tqdm progress bar using pure REST API.

Halo fix for ipython>8.17.0 - Fix for halo package compatibility with ipython versions higher than 8.17.0.

pylandau logging enhancement - Added logging mechanism to pylandau package to handle warnings instead of using print statements.

PUBLICATIONS

For more, visit InspireHep.

- Cosmic Ray Measurements Using Charge and Light Readout in a Pixelated Liquid Argon Time Projection Chamber
  The SoLAr Collaboration
  arXiv:2512.10830 [physics.ins-det] (2025)
- Opportunities and challenges to study solar neutrinos with a Q-Pix pixel readout
  The Q-Pix Collaboration
  Phys. Rev. D 112, 092009 (2025)
- First Demonstration of a Combined Light and Charge Pixel Readout on the Anode Plane of a LArTPC
  The SoLAr Collaboration
  JINST 19 (2024) 11, P11010
- Distinguishing circumstellar from stellar photometric variability in Eta Carinae
  A Damineli, E Fernández-Lajús et. al.
  Monthly Notices of the Royal Astronomy Society, 2019
- SoLAr: A novel technology for solar neutrino detection
  Guilherme Ruiz Ferreira
  NuPhys2023: Prospects in Neutrino Physics, 18-20 December 2023, London, UK
- SoLAr: Solar Neutrinos in Liquid Argon
  The SoLAr Collaboration
  Snowmass, 2021

SKILLS & PROFICIENCY

Python

Git

Arduino

Java

C++

ROOT

MS Office (Excel, VBA)

SQL

HTML

CSS

LaTeX

Mathematica

EDUCATION

PhD in Particle Physics
University of Manchester
2021 - 2025

STFC Summer School
University of Oxford
2022

MSc in Physics
University College London
2018 - 2020

BSc in Physics with Astronomy
University of São Paulo
2013 - 2018

First joint ICTP-Trieste/ICTP-SAIFR School on Particle Physics
IFT-UNESP
2018

Particle Physics Summer Course
University of São Paulo
2017

LANGUAGES

Portuguese (Native)

English (Fluent)

Spanish (Basic)